Interior Temporal Stability of Object-Space Feature Lines on a Frozen 3DGS: an Isolated Finding at Matched Precision and Density, and a Measured Precision Boundary

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***Abstract*—Line drawings rendered from per-frame image-space edge detection flicker: strokes appear, vanish, and re-form with every camera step. We report an interior temporal-stability finding for feature lines bound to a *frozen* 3D Gaussian Splatting reconstruction — static object-space 3D polylines, projected per frame with a visibility test — together with a diagnostic characterization of the precision such lines can reach. This is deliberately not a general 3D line detector: view-dependent silhouettes are out of scope by construction, exhaustive recovery of geometric creases is a ceiling we measure rather than hide, and at disocclusion boundaries our stability advantage reverses — a limit we quantify. Per-frame 2D detection remains the precision reference we trade against rather than claim to beat. What object-space binding buys is measured on a three-axis protocol that compares methods only at matched precision and matched line density, our lines pop 1.72–8.35$\times$ less per scene$\times$trajectory condition (two synthetic scenes $\times$ two trajectories, four conditions; a third trajectory appears in the stroke-level results) than the strongest 2D baseline we could construct — an EMA accumulator driven by *oracle* rigid flow, so no estimated flow can improve its inputs. The worst cell, 1.72$\times$ (the stress spline on the geometry-dense scene), breached our pre-registered 2$\times$ bar and is reported as the claim's floor rather than re-tuned; the other three conditions sit at 5.19–8.35$\times$, and the advantage over memoryless detectors is $\geq$9.8$\times$. These are stability numbers at the lines' *bounded* precision — mesh-free crease-vs-texture discriminability caps at 0.6371 AUC (below its 0.72 gate) — on n=2 synthetic scenes. A second contribution characterizes why the primitive's precision is bounded rather than patching it: coverage is capped by the frozen carrier; the missing creases carry no geometric signal even under the ground-truth mesh (AUC 0.3964); the crease-vs-texture signal *exists* in frozen DINOv2 features (AUC 0.8401/0.9044) yet collapses to 0.6371 under our best mesh-free supervision through a pre-registered 0.72 gate — precision is supervision-bound under our frozen protocol, not solved — and the obvious patch is falsified: no post-hoc discriminator filter we tested — up to an in-scene mesh oracle — reached the pipeline's own precision at matched recall without severing crease connectivity (Appendix A). Every experimental gate in the paper was frozen before its numbers existed; all outcomes, including the unfavorable ones, are reported.**

1. Introduction

Stylized and technical renderings of 3D scenes want *lines* — creases, seams, part boundaries — and they want those

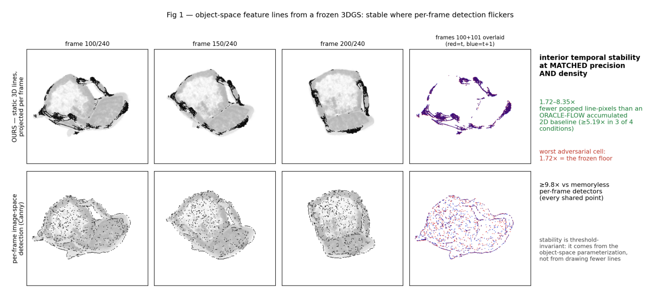


Fig. 1. Teaser: object-space strokes versus per-frame line detection, with a two-frame overlap visualizing stroke identity across a camera step.

lines to hold still. A line drawing produced by running an edge detector on every rendered frame is precise frame-by-frame but temporally incoherent: most of its strokes do not survive even one frame transition (Fig. 1). With 3D Gaussian Splatting (3DGS) [1] now a standard scene representation, we ask a narrow question: given a *frozen* 3DGS — no retraining, no densification — can feature lines be bound to the reconstruction so that they are *stable* under camera motion? We answer for the interior of the object only, and we say up front what this work is **not**: it is not a general 3D line detector — it does not attempt exhaustive recovery of dihedral creases (a ceiling we measure, not hide), it does not draw view-dependent silhouettes (all methods, ours and baselines, are evaluated interior-restricted), and its stability advantage *reverses* at disocclusion boundaries, which we quantify rather than average away.

The difficulty in evaluating such a primitive is that temporal-stability claims are cheap: a method looks stable if it draws fewer lines, easier lines, or vaguer lines. Our protocol closes these doors. Every method traces an operating curve over precision and line density, and stability is compared only at operating points shared on *both* axes; the stability statistic is pooled per line-pixel through an exact rigid-flow warp, so vanished lines are charged automatically. And because per-frame detection has an obvious repair — warp and accumulate edges over time — we build the strongest member of that family we know how to construct: an EMA accumulator given our own *oracle* rigid flow with an occlusion-aware fallback — no estimated flow can improve on its inputs, though we make no claim over accumulator designs beyond this family.

Against this ceiling, the stability finding is **1.72–8.35×** fewer popped line-pixels per condition on the two scenes (≥5.19× in three of four; the fourth — the stress spline on the geometry-dense scene — **breached the pre-registered 2× bar at 1.72× and stands as the reported floor**), and ≥9.8× against memoryless detectors at every shared point — all at the lines' bounded precision (mesh-free discriminability 0.6371 AUC; §5). The advantage is interior (1.98× at the hardest cell) and is a property of the parameterization: it is invariant to the acceptance threshold across the full sweep where measured (the chair sweep; not repeated on lego), while per-frame detectors destabilize as they sparsify.

Precision is the other half of the story, and we do not claim to have solved it. Per-frame image-space detection is the precision reference this primitive trades against: on the texture-bound scene our lines exceed the seed edge field's precision at matched density — evidence of genuine multi-view selectivity — but on the geometry-dense scene per-frame Canny stays more precise at every density we reach. Rather than patch this gap with an in-scene oracle, we characterize it: a four-act forensic bounds the achievable precision (carrier coverage ceiling; no geometric cue on the miss-set, even from the GT mesh; the separating signal exists in frozen semantic features; and it collapses without mesh supervision through a pre-registered gate), ending at a route that worked in one of the two directions tested (0.8245 chair→lego; 0.5626 reverse) — then pressure-tests the boundary itself: post-hoc filtering of the line-candidate cloud is falsified as a repair class, oracle ranking included (Appendix A). The boundary is the contribution; pretending it away is not.

Contributions. (condensed from the contribution box) 1. **An interior temporal-stability finding for object-space lines on a frozen 3DGS** — 1.72–8.35× less popping than an oracle-flow accumulated 2D baseline at matched precision and density (≥5.19× in three of four conditions; the 1.72× cell breached its pre-registered 2× bar and is the reported floor), ≥9.8× vs memoryless detection — on n=2 synthetic scenes, at precision bounded by contribution 2 (mesh-free discriminability 0.6371 AUC); per-frame 2D detection remains the precision reference we trade against, not a baseline we claim to beat in general. 2. **A measured precision boundary, with the obvious repair falsified** — the four-act characterization ending supervision-bound under our frozen protocol (signal exists 0.8401/0.9044; mesh-free 0.6371 vs a 0.72 gate; transfer 0.8245 one direction), plus a structural-impossibility result for post-hoc candidate filtering: no tested gate — mesh-free or in-scene mesh oracle, point- or chain-pooled — reaches the pipeline's precision at matched recall while preserving crease connectivity (the topological trilemma, Appendix A). Every recovery attempt we falsified is reported — a boundary with a named route forward. 3. **Pre-registered gates as method** — every gate frozen before its numbers existed and evaluated on its letter, unfavorable outcomes included. The mesh-free guarantee is a runtime and test-time property; development-time model selection used mesh-scored *validation* views, disclosed in §3.4.

2. Related Work

(Our own measurements cited here are ledger-exact from `RESULTS_MASTER.md`.)

Per-frame image-space edge detection. Classical (Canny) [2] and learned (DexiNed, TEED, PiDiNet) [3], [4], [5] detectors produce precise, dense 2D edge maps per rendered frame and are, for our purposes, the **precision reference we trade against rather than claim to beat**: on the geometry-dense scene, per-frame Canny on the same rendered inputs is more precise than our lines at every density we reach (§5.1). What per-frame detection cannot supply is identity over time — most of its strokes do not survive a single frame transition (§4.5) — and §4 shows that even granting this family an oracle-flow temporal accumulator does not close the stability gap at matched precision and density.

3D edge and curve reconstruction from multi-view edges. EMAP ("3D Neural Edge Reconstruction", CVPR 2024) [7] and its successors learn a neural **unsigned distance field to edges** from multi-view 2D edge maps and extract parametric curves from it. This line of work fuses *edges of any kind*: the field inherits whatever the 2D detector fires on, with no mechanism to distinguish a geometric crease from a printed texture boundary. Our boundary characterization (§5.2) quantifies exactly this blind spot as a representational fact rather than an implementation gap — on decal-like structure, *no* geometric cue separates the classes, **including the ground-truth mesh's own dihedral (AUC 0.3964)** — so edge-field fusion, like our carrier, is semantics-blind by construction; the separating signal is semantic (§5.3) and supervision-bound (§5.4).

SketchSplat [arXiv 2503.14786] [8] optimizes parametric 3D edges differentiably against multi-view 2D edge maps, and motivates its design in part by the observation that detectors mis-fire on shading textures and produce view-inconsistent edge shifts. Two of its ideas map onto our study with opposite outcomes. Its **curve-level aggregation** — deciding at the curve, not the pixel — transfers well as a *read*: our chain-level reads consistently strengthen per-point signals (§5.3) — though Appendix A bounds what chain-level *gating* of a fixed candidate cloud can deliver. Its **view-consistency premise** — that texture mis-fires can be filtered because they are view-inconsistent — **does not transfer to our setting**: on rigid, static scenes, printed *albedo* texture edges are fixed surface loci and reproject almost as consistently as true creases (multi-view consistency 0.870 vs 0.937; §5.2), so a consistency filter cannot carry the crease-vs-texture decision here — whereas the specular and shading-induced mis-fires that motivate the premise are a different error population that our scenes barely contain. We read this as a domain statement, not a criticism of SketchSplat's results on its own benchmarks. A fair question is why no EMAP/SketchSplat-style static-curve output appears as a *stability* baseline: any static object-space curve set shares our by-construction stability — which is precisely why our contribution is stated as a *measurement* of what staticness is worth against the strongest dynamic family at matched precision and density, a quantity no static-curve paper reports, rather than a ranking among static methods. Between static primitives the discriminating axes are precision, coverage and density — which our §5 characterization addresses at the representation level (its semantic-blindness result is representation-independent — even the GT mesh's dihedral scores 0.3964 — and so applies to any edge-seeded method, theirs included; the coverage ceiling, by contrast, is measured for our carrier only) — and our frozen evaluation did not extend to third-party curve sets. We state this as an evaluation boundary, not an implied superiority.

Temporal coherence in stylized rendering. Coherent line drawing and stylization classically bind strokes to object space or propagate them with flow — the standard answers to flicker. Our contribution to this literature is not the idea of object-space binding but its **measurement discipline**: stability compared only at matched precision AND matched line density, against an accumulator granted oracle flow, with pre-registered gates and the failure envelope (disocclusion reversal, 1.72× floor) reported as part of the claim.

Feature lines from 3D representations. Mesh-based feature-line extraction (ridges, valleys, suggestive contours) presumes trustworthy surface geometry. A frozen 3DGS offers no such thing at fine scales; §5's four-act arc measures how far post-hoc extraction can go on such a carrier and where it stops — the ceiling (§5.1), the geometric blindness (§5.2), and the supervision bound (§5.4).

3. Method

(Protocol constants are the ledger's; the pipeline is described structurally — its operating points are swept in §4, not fixed here.)

3.1 The invariant, first

One rule governs everything downstream, and we state it before the pipeline: **the ground-truth mesh never enters the method path** — a runtime and test-time guarantee whose development-time boundary we disclose in §3.4. It is confined to a single evaluation module (`mesh_oracle.py`) that labels held-out GT crease points and scores precision/recall — nothing the pipeline computes, at any stage, reads it (AST-verified per phase). Every signal the method consumes comes from the frozen 3DGS itself (gaussian positions, opacities, rendered depth/normal/alpha buffers) and from rendered images of the 80 training views; the 10 test views and all test trajectories are held out from every fitting step.

3.2 A coarse object-space carrier, corrected by image-space evidence

The design splits the labor between two unequally reliable sources. The **carrier** is coarse and 3D: the frozen gaussian cloud (de-floatered by opacity and a k-NN spacing test) anchors *where in space* feature lines may live and supplies visibility — but its fine-scale geometry is untrustworthy (§5.2), so it is never asked to localize an edge. The **corrector** is precise and 2D: per-view **Canny** edge maps of the training photographs (unblurred, hysteresis 50/150), converted to sub-pixel **distance-transform (DT) fields**, one per training view. A learned detector enters only through the *seed ranking*, never the pull field: seeds are ranked by a multi-view photometric score whose edge source is a blurred Canny union in the shipped configuration and, in the matched-precision configurations of §4.1–4.4, **TEED** — a frozen, zero-shot, BIPED-trained detector of 58,910 parameters, chosen among frozen detectors on validation views and never fine-tuned — NMS-thinned and thresholded at 0.5. Short oriented 3D segments ("linelets" — a position, a unit tangent, and a half-length, one per kept carrier seed, initialized from the carrier's local structure) are then optimized by gradient descent so that samples along their projections descend these DT fields *in many views simultaneously* — the multi-view DT pull, run for a fixed iteration budget with a per-step displacement cap. A linelet that sits on a real 3D feature line finds a consensus position that is on-edge in most views at once; one seeded on a single-view artifact cannot. Robust per-view statistics accumulated during the pull provide the acceptance criterion — a linelet is kept iff it is visible in $\geq$3 pull views with inlier fraction $\geq$0.50 and median on-edge residual $\leq$1.5 px (frozen defaults) — and a single keep-fraction knob over the carrier's seed ranking traces the operating curve that §4 sweeps.

3.3 From linelets to renderable, stable strokes

Accepted linelets are non-maximum-suppressed in 3D and greedily chained into polylines using tangent and collinearity consistency, yielding a static set of object-space 3D strokes — computed **once** for the frozen scene. Rendering at a novel camera is then projection plus occlusion: each vertex is tested against the 3DGS z-buffer (a 3×3-min window with a small relative tolerance), strokes are split into visible runs, and runs are rasterized at one pixel. Temporal stability is not enforced by any tracking, smoothing, or hysteresis — there is nothing to track, because the primitive is static in object space; §4 measures what this buys and §4.4 what it costs (the visibility split is exactly why our advantage reverses at disocclusion boundaries).

3.4 Where the invariant stops: hyperparameter provenance

The AST-checked invariant of §3.1 is a statement about the *runtime* method path. We extend it, honestly, to the experimenter: the frozen view split (80 train / 10 validation / 10 test) governs all selection — pipeline constants (detector and threshold, de-floater cutoffs, acceptance defaults, chaining tolerances) were chosen during development using scores on *validation* views, and those development-time scores did use the evaluation oracle; that is a disclosure, not a violation, because every number this paper reports is computed on the 10 held-out **test** views and on test-anchored trajectories that no selection step ever consumed, under gates frozen before the numbers existed. What pre-registration covers here is the reported evaluation, not the existence of a development loop — and with n=2 scenes there is no held-out *scene*, a scope limit §6 owns. This loop is also why §5.4 matters beyond the discriminator: deploying on a mesh-less scene would require mesh-free selection, whose strongest tested form §5.4 bounds; the tested mitigation is cross-scene transfer of development-time choices (0.8245 in the one direction that worked).

Frozen constants (committed defaults, printed for reproduction). De-floater: opacity > 0.1 and k-NN (k=8) mean spacing < 3× its median. Pull field: Canny, no blur, hysteresis 50/150, distance-transformed (every configuration). Seed ranking: Canny-sourced (blurred two-scale union, σ = 2.0 and 2.5, thresholds 100/200 and 75/150) for §5.1 and §4.5; TEED-sourced (native scale, NMS-thinned, threshold 0.5) for the sweeps of §4.1–4.4. DT pull: 100 gradient steps, learning rate 0.35, per-step displacement cap 5.0 px. Acceptance: visible in $\geq$3 pull views, inlier fraction $\geq$0.50, median residual $\leq$1.5 px. Chaining: 3D NMS radius 1.0× local spacing, k=10 neighbors, tangent cosine $\geq$0.60, collinearity cosine $\geq$0.50, gap $\leq$4.0× spacing, $\geq$3 nodes per stroke. Visibility: 3×3-min z-buffer window, relative tolerance 0.02. Seed construction and half-length initialization follow the released implementation (`src/seeds.py`, `src/dt_pull.py`), which we release.

3.5 What the method does not contain

No mesh in the runtime path (§3.1, provenance in §3.4). No per-frame optimization. No temporal filter. No learned component beyond the frozen zero-shot 2D detector that ranks seeds in the §4.1–4.4 configurations (the pull field is Canny throughout). The multi-view DexiNed triangulation and the DINOv2 probe of §5 and Appendix A are *diagnostic instruments*, not stages of the shipped generator: no triangulated candidate becomes a stroke and no discriminator threshold is deployed; the shipped generator is exactly §3.2–3.3. The evaluation protocol (matched precision-and-density comparison, pooled warp metric, interior restriction) is specified with the experiments in §4.1, because it is shared by every baseline rather than being part of the method.

4. Interior stability at matched precision and density

(Contribution A, primary. All numbers from `RESULTS_MASTER.md`; figures Fig. 2–5, Tab. 1–2.)

4.1 Protocol: three axes, one dominance rule

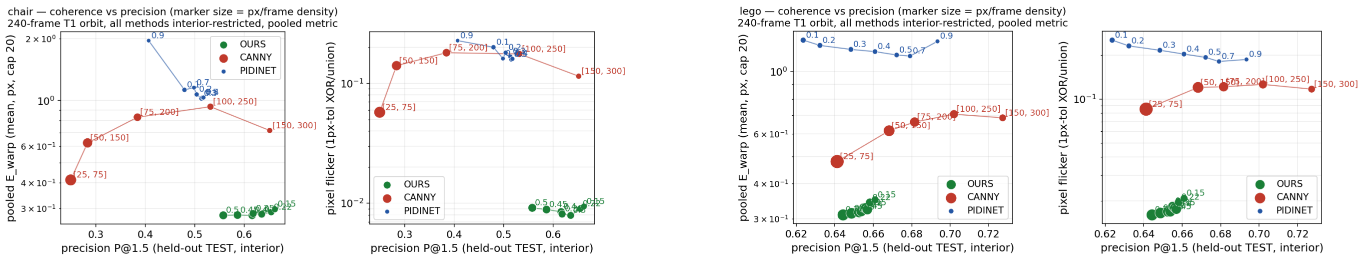
A temporal-stability claim for a line primitive is easy to fake and easy to dismiss: a method can look stable by drawing fewer lines or blurrier lines. Our protocol closes those doors — precision and pixel-density are matched jointly — but we name what it does not close: matched density is not matched *coverage* (§5.1 shows creases with no carrier at all, which our lines cannot ink at any density), so the comparison certifies stability of what is drawn, not equivalence of what is drawable. Every method — ours and every baseline — is swept over its acceptance threshold to trace an operating curve over three axes: precision (P@1.5 against held-out GT creases), line density (rendered line-pixels per frame), and a pixel-pooled stability statistic. Stability is compared **only at shared operating points**: a baseline point counts iff some point of ours matches or beats it on *both* precision and density; the reported advantage is the *minimum* ratio over all such dominating points. The stability statistic itself is pooled per pixel, not per line: every ON line-pixel of frame *t* is forward-warped by the exact rigid flow (rendered depth plus the two camera poses — the same operator for every method), and the distance transform of frame *t+1*'s line mask is read at the landing pixel; distances are pooled over all 239 transitions of a 240-frame trajectory. There is no matching step and no per-line normalization, so a vanished line is charged automatically, and a sparse "sticky" line set cannot be flattered. All methods are restricted to the object interior (α>0.5, eroded 2 px), which removes the silhouette warp-drop confound at source for everyone (warp-drop $\leq$0.1 % throughout).

4.2 Against memoryless per-frame detection

At every shared operating point on both scenes, our projected object-space lines flicker $\geq$**9.8**× less than per-frame Canny and PiDiNet (pixel flicker, 1 px-tolerant XOR/union; Fig. 2). The more telling observation is *how* the two families traverse their operating curves: our pooled pop-rate is **invariant to the acceptance threshold** (P(d>2 px) stays within 0.0042–0.0049 across the entire sweep on chair) — stability is a property of the object-space parameterization, not of which lines are kept — while the per-frame detectors *destabilize* as they sparsify (PiDiNet at thr 0.9 pops 3× more than at 0.1). One pre-registered gate in this comparison failed, and we report it rather than re-tune it: the pooled-*mean* statistic's advantage drops to **2.42**× at the sparsest high-precision Canny point, below our frozen 3× bar. The dissection (Tab. 2) shows why the statistic, not the stability, collapses: our lines are static 3D polylines whose true inter-frame drift is zero, so their pooled mean sits at the shared ~0.28 px rasterization/warp quantization floor (p95 = 1.00 px, threshold-invariant), and a mean over a floor-dominated distribution compresses ratios at sub-pixel motion. The floor-free statistics at the *same* point read 12.8× (pop) and 12.2× (flicker). We quote pop and flicker as primary and disclose the mean-statistic collapse.

4.3 Against an oracle-flow temporally-accumulated ceiling

The strongest objection to §4.2 is that per-frame detection is a strawman: a reviewer's baseline would warp and accumulate edges over time. We therefore built the strongest member of that family we know how to construct — an EMA accumulator (A_t = α·warp(A_{t−1}) + (1−α)·E_t, rethresholded, α up to 0.85) driven by the **exact rigid flow** with an occlusion-aware fallback, i.e. an oracle upper bound on every estimated-flow variant — and swept it on the same three axes (Fig. 3); the oracle claim covers the flow *input* only — we make no claim over accumulator designs beyond this EMA family. Accumulation genuinely helps the 2D baselines. It does not



(a) chair

(b) lego

Fig. 2. PARETO-1 precision/line-density operating frontiers: (a) chair, (b) lego.

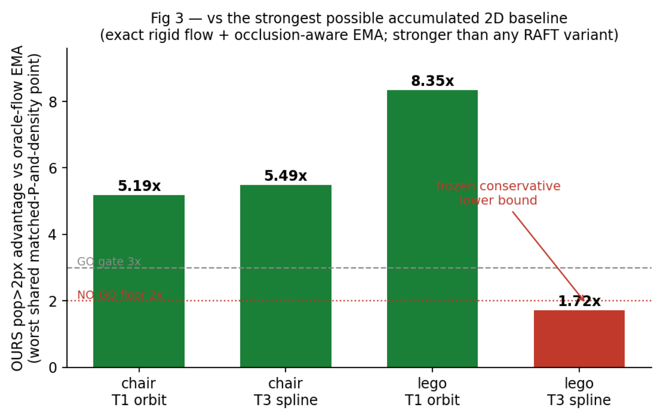


Fig. 3. Per-condition worst-case stability advantage against the oracle-flow accumulated baseline; the gate-breaching lego×T3 cell is flagged as the reported floor.

close the gap on the pooled pop-rate P(d>2 px) — the floor-free statistic §4.2 motivates: the worst shared advantage per condition is **5.19**× (chair·orbit), **5.49**× (chair·spline), **8.35**× (lego·orbit), and **1.72**× (lego·spline); §4.4 decomposes where this advantage lives and how much of the baseline's residual is its own accumulator drift. The shared operating points sit at P@1.5 0.28–0.59 on chair (9 and 8 of 21 baseline points shared) and 0.62–0.64 on lego (3 and 5 of 21) — on lego the baseline's highest-precision configurations are not dominated by any of our points and are therefore outside the comparison, a truncation the dominance rule imposes by construction and we report rather than hide. The last cell breaches our frozen 2× bar and we keep it as the headline of this subsection rather than a footnote: **1.72× is the reported floor of the claim — a breached pre-registered gate, not a designed margin** — measured against a baseline whose flow input no practical system can exceed, in the single condition that maximizes occlusion flux.

4.4 The operational envelope

Where does the advantage live, and where does it stop? Decomposing the worst cell by region (Fig. 4): the advantage is **interior** — pop-rate 0.0214 vs the accumulator's 0.0425 (**1.98**×) over the 93–97 % of line-pixels away from occlusion boundaries — and it *reverses* inside disocclusion regions, where our rate (0.407) is worse than the baseline's (0.300): our visibility test splits chains at occlusion boundaries and the run endpoints shift. We disclose this rather than a reviewer discovering it. A pre-registered mechanism gate — "≥60 % of the accumulator's residual popping lies in disocclusion regions" — came back **33.3 %** (NO-GO): the accumulator's residual is diffuse interior EMA drift, not a disocclusion-correspondence failure, so this paper makes **no** mechanism claim about *why* 2D accumulation trails; the bound is empirical. The envelope is nonetheless predictable: the single sub-2× cell is exactly the condition (micro-relief geometry × non-uniform multi-axis motion) where per-frame line-pixel turnover is dominated by occlusion-boundary churn.

4.5 Stroke-level corroboration

The pixel-pooled results above are corroborated by the independently built stroke-level harness (Tab. 1, Fig. 5): matched-stroke E_warp ratios of **3.38–21.62**× across 2 scenes × 3 trajectories against per-frame TEED (scorer and thresholds hash-frozen before any number existed), Fréchet ratios of 2.43–29.92× against per-frame Canny, and — most legibly — stroke *survival*: object-space strokes persist for 37–183 frames on average where per-frame strokes persist for 1.0–1.5 (P(lifetime>32): 0.29–0.83 vs 0.005–0.009). Our stroke residual falls in proportion to per-frame motion, i.e. it is warp-resampling error and nothing else; the per-frame baselines saturate at a motion-independent popping floor. Fig. 9 shows that comparison directly across the whole frame sweep together with both of its confound controls — the sparsity check and the silhouette warp-drop control — and Tab. 5 collects the per-scene accuracy and coherence figures in one place. These stroke-level harnesses predate the accumulated baseline and compare against memoryless detection only; the oracle accumulator exists only in the pixel-pooled protocol, so the third trajectory is covered at stroke level but not against accumulation.

(Scope reminder for the section header: frozen 3DGS, static NeRF-synthetic scenes, known poses; see §6. The precision these curves operate at is itself bounded — §5 characterizes that boundary.)

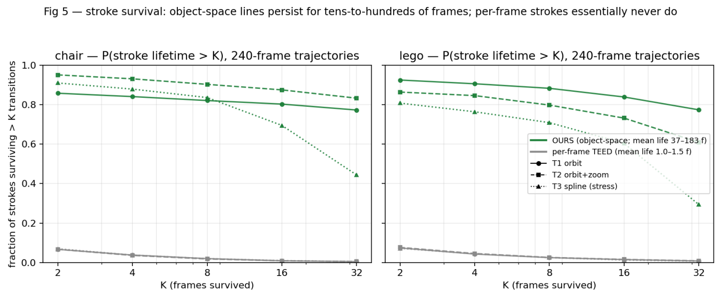


Fig. 5. Stroke-survival curves P(life $>$ K) for the six scene$\times$trajectory conditions: object-space strokes persist for tens-to-hundreds of frames; per-frame strokes essentially never do.

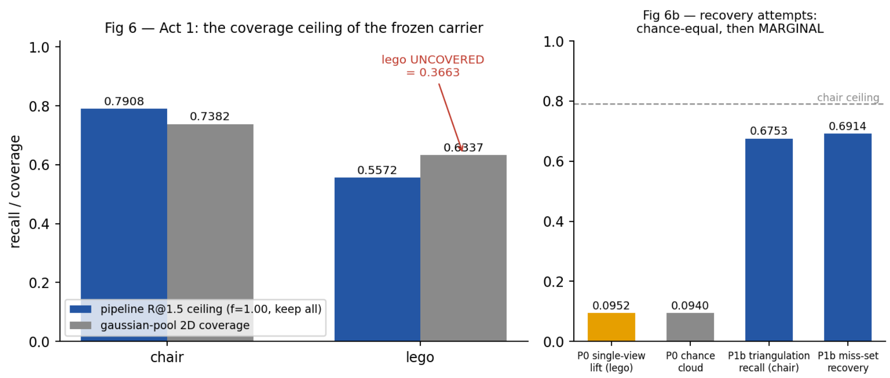


Fig. 6. Act 1: the carrier coverage ceiling, and the recovery ladder from single-view lifting to multi-view triangulation.

5. The Precision Boundary: A Four-Act Characterization

(Contribution B, diagnostic. All numbers from `RESULTS_MASTER.md`*; Fig. 6–8, Tab. 3–4. Throughout, the GT mesh labels and scores only — it never enters the method path.)*

5.1 Act 1 — the ceiling exists, and what it does and does not mean

The primitive of §4 operates at a bounded precision, and the bound begins with coverage: re-ranking the frozen gaussian pool — keeping *everything* — caps pipeline recall at R@1.5 = 0.7908 (chair) / 0.5572 (lego); the pool's own 2D point coverage is 0.7382 / 0.6337 — on chair the pipeline's recall exceeds point coverage because rasterized segments interpolate *between* carriers, so the binding form of the ceiling is lego's, where 0.3663 of visible GT crease points have no carrier within 1.5 px at all and interpolation cannot manufacture one (Fig. 6).

Both figures are stated at the crease definition we use throughout — mesh edges whose dihedral angle exceeds a fixed threshold — and lego is unusually sensitive to that choice, so we report the sensitivity rather than leave it implicit. lego's mesh contains a single large family of edges lying at exactly the threshold angle, which the threshold splits near-evenly through the asset's six-decimal vertex quantisation; nudging the threshold by a twentieth of a degree therefore discards nearly half of lego's crease pixels and raises the measured ceiling materially, after which it plateaus. This is a property of the asset, not of the reconstruction or of the extractor, and it does not describe a better pipeline: precision over the identical rasterised segments falls by more than half across the same range, F-score is flat-to-declining, and the joint operating gate is met at no threshold on either scene. chair, which carries no such family, is unaffected. The ceiling conclusion is therefore robust to the threshold even though the individual number is not; the per-threshold values are tabulated in the results ledger.

The crease set is also a topology-selected subset of the geometric $\geq 30°$ edges — **37 % (lego) / 43 % (chair) / 31 % (ficus)**, Appendix A.5 — and every precision/recall number here is relative to it. Some of those uncovered creases are flat decals with literally zero geometric footprint.

This admission invites a specific attack, so we answer it here rather than in a rebuttal: *if the lines live on a baked radiance carrier, is §4's stability just the passive reprojection of 2D edges — an inherited artifact that this ceiling proves?* Not on the evidence — though the evidence is single-scene, as §6 states plainly. On the texture-bound scene, the carrier surface under a high-contrast fabric edge is indistinguishable from the carrier under a crease — on this population the fabric-bound rows of Act 2 apply directly (the albedo-step gate leaks, 0.1235 vs 0.1211, and multi-view consistency does not separate, 0.870 vs 0.937), and the geometric cues of Tab. 3 are at or below chance on the decisive decal population — yet our extracted set is substantially **more precise than the photometric edge field itself** — P@1.5 0.662 at 4,947 px/frame vs per-frame Canny 0.532 at 5,973 px/frame, and, in the dominance-consistent direction (ours denser AND more precise), 0.636 at 7,942 px/frame vs the same 0.532 (Fig. 2): the pipeline actively **suppresses** texture edges that enjoy the same carrier support and the same image contrast as the creases it keeps. That is genuine multi-view selectivity performed by the object-space aggregation — work the carrier could not have done for us (Act 2) and the 2D edge field did not do for us (Fig. 2) — not passive reprojection. The converse case gets equal prominence: on lego, where strong image edges largely *are* creases, per-frame Canny is more precise than our lines at every density; there our advantage is stability only. The ceiling, then, bounds *which* creases the primitive can carry — it says nothing against *how stably* it carries them, which is §4's claim and survives at matched precision and density.

5.2 Act 2 — no geometric cue separates the miss-set (K_geom ≈ 0)

Could a better geometric gate recover the uncovered creases or reject the texture? On the decisive population (lego decals vs true creases), no geometric channel is usable (Tab. 3; several score *below* chance — decals locally out-score creases — so even sign-flipped they reverse the intended semantics): 2DGS surfel dihedral 0.4110, rendered-normal ribbons 0.3307/0.3875, and — the controlling result — the **GT mesh's own dihedral, 0.3964**, with normal-dispersion medians 44.52° vs 44.84°: given *perfect* geometry, the two classes differ by 0.32°. The low-level photometric escape routes close the same way: the SH-DC albedo-step gate leaks (fabric p50 0.1235 vs crease 0.1211), and multi-view consistency does not separate (texture 0.870 vs crease 0.937 — printed patterns are rigid surface loci too). Coverage recovery fares no better: single-view photometric lifting equals its chance control (R_miss 0.0952 vs 0.0940), and multi-view triangulation, though a genuine localization fix, stays detector-bound at MARGINAL (recall 0.6753 / miss-set recovery 0.6914; Fig. 6b).

5.3 Act 3 — the separating signal exists

The missing discriminator is not hiding in geometry; it is semantic. A linear probe on frozen zero-shot DINOv2 features [6] separates crease from texture at AUC **0.8401 / 0.9044** (chair / lego, held-out spatial split), 0.8205 / 0.8913 under the leakage-guarded chain-level read, against ≈0.71 photometric and ≈0.65 geometric probe baselines on the candidate population — a different population and feature set from §5.2's decal test, which is why a nonzero geometric number here does not contradict K_geom≈0 there (Fig. 7). Read honestly, the probe recognizes *which surface* a point lies on — fabric field vs piping, stud field vs decal — a surface-identity readout rather than an edge-type detector. The signal the primitive needs exists, in features every modern pipeline already has — though Act 4 shows that reading it out is supervision-bound.

5.4 Act 4 — and it is supervision-bound under our frozen protocol

The signal exists (0.8401/0.9044) — but it collapses under mesh-free supervision (Fig. 8): trained on our best mesh-free pseudo-labels (physics-frozen geometric-photometric votes, and a fully self-supervised clustering variant), the same probe falls to **0.6371** on chair through a pre-registered 0.72 gate (on lego 0.9046 — the ceiling as recomputed in this study; 0.9044 in Act 3's split — falls to 0.6569), beneath even its photometric baseline: the best mesh-free number *anywhere* is the chair photometric probe at 0.7326 — touching the gate on one scene, 0.5637 on the other — while the semantic probe collapses on both. The failure mode is instructive: the pseudo-labels' errors are systematic, not noisy, and the richer representation learns them the more faithfully — zero denoising. One route stays open, and we measured it: the mesh-supervised probe transfers chair→lego at **0.8245**, nearly matching lego's in-scene ceiling, though not in reverse (0.5626). Precision is therefore **supervision-bound under our frozen protocol** — with labeled training scenes as the named, tested-in-one-direction path forward. Appendix A closes the loop at deployment granularity: used as an actual gate on the line-candidate cloud, the deployment-legal direction of that transfer (0.5626, chance) leaves precision untouched, and even an in-scene mesh oracle cannot filter the cloud to the pipeline's precision without severing crease connectivity.

5.5 What the boundary buys

This characterization is what licenses §4's scope. The arc — coverage bounded by the carrier (5.1), the miss-set geometrically invisible (5.2), the discriminator semantic and extant (5.3), its mesh-free readout falsified through a frozen gate (5.4) — explains *why* we ship a stability primitive at a measured precision rather than patching precision with an in-scene oracle: every patch we tested either equals chance, stays detector-bound, requires supervision the method path is not allowed to touch, or — when granted that supervision as an explicit oracle — fragments the very chains it is meant to purify (Appendix A). The boundary is measured, disclosed, and has exactly one open door: labeled scenes feeding a topology-aware construction, not a post-hoc filter — and a learned 3D edge field is not a second door (Appendix A.4).

6. Limitations

(All numbers from `RESULTS_MASTER.md`. Nothing here is softened; several of these limits were established by our own pre-registered gates coming back negative.)

In-vitro scope. Everything in this paper is measured on frozen 3DGS reconstructions of two static NeRF-synthetic scenes [9] with known poses and rendered depth — an in-vitro geometric characterization, owned as such rather than patched. Every baseline shares the same perfect-geometry assumptions (the accumulated baseline is *given* our exact rigid flow), so the comparisons are internally fair, but n=2 synthetic scenes with perfect geometry is the breadth we have; real captures, estimated poses, and more scenes are future work, not implied results.

The selectivity evidence is single-scene. The §5.1 demonstration that our pipeline actively suppresses texture edges — higher precision than its own seed edge field at matched density — holds on chair only. On lego, per-frame Canny is *more precise than our lines at every density we reach*; our advantage there is stability alone. Selectivity is demonstrated on the texture-bound scene and not contradicted on the geometry-dense one. The stability claim itself does not lean on this: §4.1's dominance rule compares stability only at operating points the baselines actually share on both precision and density, so §4's ratios survive on lego with no selectivity assumption at all.

The temporal advantage is interior-only, and reverses at disocclusions. Decomposing the hardest cell (§4.4): the entire advantage is interior EMA-drift suppression — pop-rate 0.0214 vs the oracle accumulator's 0.0425, **1.98**×, over the 93–97 % of line pixels away from occlusion boundaries — while inside disocclusion regions **our method is worse** (0.407 vs 0.300): visibility-culled chain runs split and their endpoints shift. Our pre-registered mechanism gate ("$\geq$60 % of the accumulator's residual lies in disocclusion regions") came back **33.3 %**, a NO-GO, so we claim **no** disocclusion-correspondence mechanism for why 2D accumulation trails; the temporal bound is empirical.

Precision is not solved — it is supervision-bound. The crease-vs-texture signal exists in frozen DINOv2 features (0.8401/0.9044 with mesh labels) but collapsed to **0.6371** under our best mesh-free supervision, through a pre-registered 0.72 gate (NO-GO). The one measured route forward — cross-scene transfer of a mesh-supervised probe — works in one direction of the two tested (0.8245 chair→lego, 0.5626 reverse). Appendix A sharpens this limit at deployment granularity: gating the line-candidate cloud with the deployment-legal transfer probe converts nothing (precision 0.3216 vs 0.3189 ungated, at matched recall), and even an in-scene mesh-oracle ranking cannot reach precision 0.71, baseline recall, and crease connectivity 0.90 simultaneously — post-hoc filtering is a falsified repair class (cloud-level probes: chair only, n=1; Appendix A). To pre-empt the reading that this falsification merely benchmarked a pathological scene: chair is the *selected* texture-stress adversary (Canny edge purity 0.28 — most of its edge field is not crease), the two barriers are mechanistic rather than chair-specific (a supervision gap that is transfer-asymmetric by measurement, and bridge-point severing that follows from thresholding any fixed candidate set), and extending the falsification to more scenes is named future work, not an implied result. Any deployment needing crease-level precision on textured surfaces currently needs labeled scenes — and, per Appendix A, a topology-aware construction rather than a filter to spend them on.

A learned edge field would not lift the ceiling either. Its closed-form upper bound fails a pre-registered gate on both scenes (lego AUC **0.698**, recall **0.000** at 85 % precision; chair **0.859 / 0.000**) — the detector's ≈5 px response tail, not missing signal (Appendix A.4).

Geometry cannot rescue it (K_geom ≈ 0). This is not an artifact of our reconstruction: even the **GT mesh's own dihedral scores AUC 0.3964** on crease-vs-decal, with normal-dispersion class medians 0.32° apart. On decal-like structure there is no geometric signal to find, for any method that seeds on geometry.

Coverage is ceiling-bound. Re-ranking the frozen carrier caps recall at R@1.5 = **0.7908 (chair) / 0.5572 (lego)**; on lego **0.3663** of visible GT crease points have no carrier within 1.5 px at all (flat decals prominent among them). Our lines cannot draw what the carrier never represented, and §5.2 shows the recovery attempts we falsified.

Further disclosed limits, collected. Four limits disclosed elsewhere belong in this list too: all pipeline constants were selected during development with mesh-scored validation views (§3.4); the pooled-*mean* stability statistic failed its own 3× gate at one point (2.42×, §4.2); no third-party static-curve method was compared (§2, an evaluation boundary); and the shared operating points on lego sit at P@1.5 ≈ 0.63 with the baseline's highest-precision configurations excluded by the dominance rule (§4.3).

Evaluation dependency. GT supervision (crease labels, precision/recall scoring) comes from the mesh, confined to `mesh_oracle.py` and the eval scripts — the method path never imports it (AST-verified per phase). The flip side of this hygiene: our quantitative evaluation is only available where GT meshes exist, which is part of why the study is in-vitro.

7. Conclusion

We set out to extract clean, temporally stable 3D feature lines from a frozen 3DGS, and we report exactly what that produced — two contributions, each scoped to what was measured.

A stability finding. Object-space feature lines whose rendered strokes are **1.72–8.35**× more temporally stable per condition than an oracle-flow temporally accumulated 2D baseline — **5.19× or better in three of four scene×trajectory conditions; the 1.72× stress-spline cell breached its pre-registered 2× bar and stands as the reported floor** — and $\geq$9.8× more stable than memoryless per-frame detection, at matched precision *and* matched line density, on n=2 synthetic scenes and at precision bounded by the second contribution (mesh-free discriminability 0.6371 AUC). The stability is invariant to the acceptance threshold across the chair sweep (not repeated on lego), consistent with it being a property of the object-space parameterization rather than of line selection. The claim ships with its measured envelope: the advantage is interior (1.98×), reverses inside disocclusion regions, and is valid for frozen reconstructions of static scenes with known poses.

A boundary forensics. The primitive's precision is *not* solved — the four-act characterization of why is itself a contribution: coverage is capped by the frozen carrier (0.7908/0.5572); the missing creases carry no geometric signal even under the GT mesh (AUC 0.3964); the discriminating signal exists in frozen semantic features (0.8401/0.9044) — and collapses under mesh-free supervision through a pre-registered gate (0.6371 vs a 0.72 bar), leaving cross-scene transfer (0.8245, one direction) as the tested route forward — and the patch family itself is falsified: no tested post-hoc filter of the candidate cloud — mesh-free or oracle-ranked, point- or chain-pooled — reached the pipeline's precision at matched recall with crease connectivity intact (the topological trilemma, Appendix A). Precision on textured surfaces is supervision-bound under our frozen protocol; we ship the boundary, measured, rather than a patch.

Methodologically, every gate in this study was frozen before its numbers existed and evaluated on its letter (the full ledger is Tab. 4); the unfavorable outcomes — including the two failed stability gates and the supervision-bound NO-GO — are reported with the same prominence as the favorable ones, because several of this paper's most useful sentences (the quantization floor, the disocclusion reversal, the oracle-baseline gap) exist only because a failed gate was dissected instead of defended. We believe the resulting object — a stability primitive with a measured boundary — is more useful to build on than either an undisclosed-limit system or an unmeasured negative.

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SUPPLEMENTARY FIGURES AND TABLES

Referenced from the main text; numbering matches the in-text references.

TABLE 1
MATCHED-STROKE E-WARP RATIOS ACROSS SCENES, TRAJECTORIES AND DETECTORS.

Tab 1 — stroke-level temporal ratios (banked; baseline / OURS, higher = steadier)

condition	E_warp ratio (vs per-frame TEED)	condition	Fréchet ratio (vs per-frame Canny)	P_pop ratio
chair · T1 orbit	20.44×	chair · 30f	4.19×	8.52×
chair · T2 orbit+zoom	21.62×	chair · 240f	29.92×	11.35×
chair · T3 spline	6.49×	lego · 30f	2.43×	3.44×
lego · T1 orbit	10.38×	lego · 240f	14.03×	11.49×
lego · T2 orbit+zoom	10.61×			
lego · T3 spline (worst)	3.38×			

TABLE 2
FLOOR ANATOMY OF THE POOLED-MEAN STATISTIC.

Tab 2 — the PARETO-1 mean-statistic FAIL, dissected (chair, Canny 150/300 vs OURS f=0.15)

statistic (pooled over all warped line px)	advantage	frozen gate	reading
pooled-MEAN E_warp	2.42×	≥3× — FAIL	floor-compressed: OURS sits at the ~0.28 px raster/warp floor (p95 = 1.00 px)
pop-rate P(d>2 px)	12.8×	—	floor-free
pop-rate P(d>3 px)	15.4×	—	floor-free
pixel flicker (1 px tol)	12.2×	—	floor-free

TABLE 3
K_{GEOM}: GEOMETRIC CUES ON THE MISS-SET.

Tab 3 — no geometric or low-level cue separates crease from texture on the miss-set (lego decals)

cue	AUC / statistic	verdict
2DGS surfel dihedral (3D)	0.4110	≤ chance
2DGS rendered-normal ribbon	0.3307	≤ chance
vanilla-3DGS normal ribbon	0.3875	≤ chance
GT-MESH dihedral (perfect geometry)	0.3964	≤ chance
GT-mesh normal dispersion	0.4675 (medians 44.52° vs 44.84°)	0.32° apart
SH-DC albedo step	fabric p50 0.1235 vs crease 0.1211	gate LEAKS
multi-view consistency	texture 0.870 vs crease 0.937	non-separating

TABLE 4
THE FROZEN-GATE LEDGER: EVERY PRE-REGISTERED GATE, ITS BAR, ITS MEASURED VALUE, AND ITS DISPOSITION.

Tab 4 — every pre-registered gate, evaluated on its letter (none re-tuned)

gate	bar	measured	disposition
PARETO-1 pooled-mean ≥3× everywhere	3×	2.42×	FAIL; floor mechanism measured, claim carried by pop/flicker
PARETO-2 vs oracle-flow EMA	2× floor	1.72×	NO-GO on the letter; frozen conservative lower bound
PARETO-3 residual in disocclusion ≥60 %	60/40 %	33.3 %	NO-GO; no mechanism sentence
Phase 0 single-view coverage R_miss	0.50/0.35	0.095 (≈chance)	NO-GO; direction killed
Phase 1b triangulation recall > ceiling	0.79	0.6753	MARGINAL; localization fix banked
Phase 1d mesh-free discriminator	0.78/0.72	0.6371	NO-GO; supervision-bound
DIAG2DGS dihedral gate	0.80	0.4110	FAIL; K_geom≈0 established

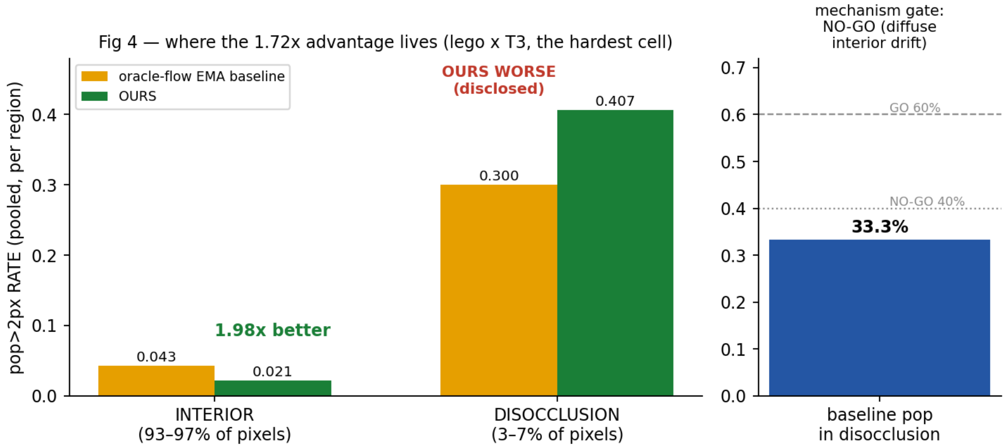

Fig. 4. Disocclusion decomposition of the hardest cell: the advantage is interior, reverses inside disocclusion regions, and the mechanism gate lands NO-GO.

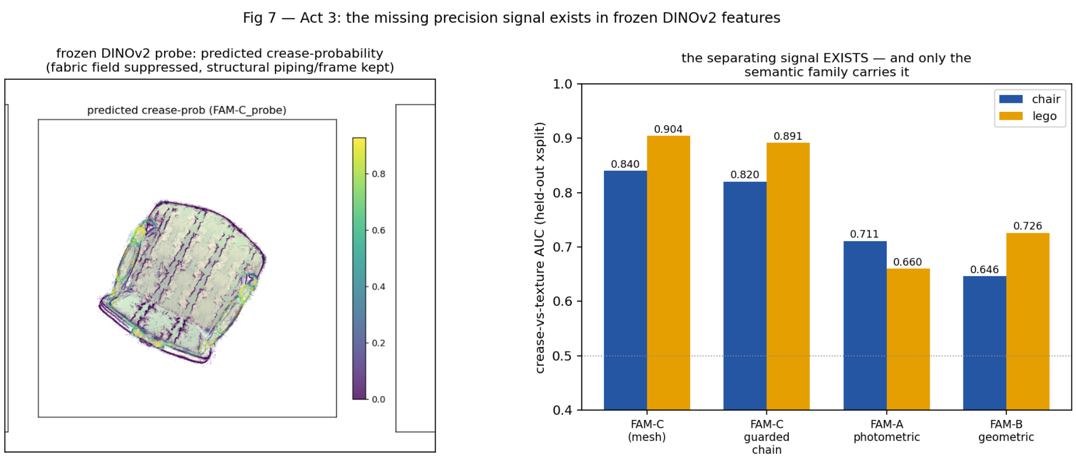

Fig. 7. Act 3: frozen-DINOv2 crease-probability read-out and probe AUCs.

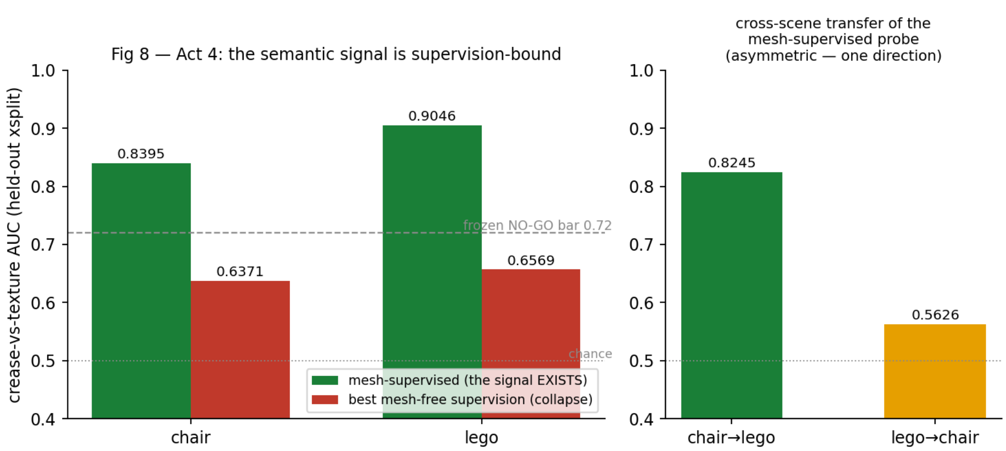

Fig. 8. Act 4: the mesh-free supervision collapse, and the asymmetric cross-scene transfer. The mesh-supervised bars show the ceilings as recomputed in the falsification study; they agree with Act 3's split values quoted in the text to the third decimal.

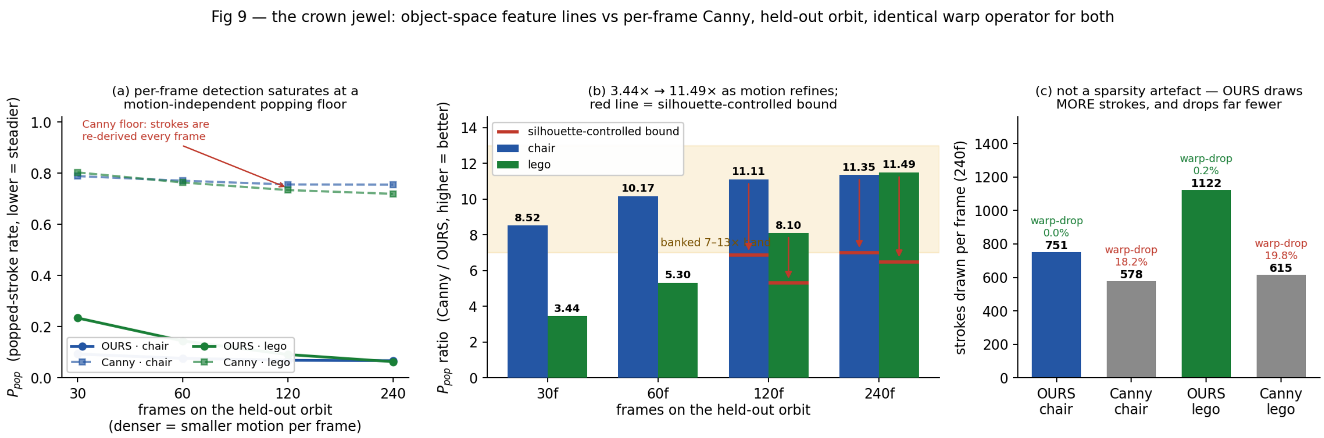

Fig. 9. Object-space feature lines versus per-frame Canny on a held-out orbit, under an identical forward-warp operator for both pipelines. Left: per-frame detection saturates at a motion-independent popping floor while object-space strokes fall in proportion to the per-frame motion. Centre: the popping-rate ratio over the frame sweep, with the silhouette-controlled bound marked in red. Right: both confounds controlled — the object-space pipeline draws more strokes per frame, not fewer, and almost none of them fail to warp.

TABLE 5

Per-scene held-out summary: segment-raster precision and recall, popping rate for both pipelines, and the popping and Fréchet ratios. Ficus is excluded by scene scoping rather than omitted — too little of its surface lies far enough from a silhouette for crease-versus-flat to be well posed — and no ficus result was ever run.

Tab 5 — per-scene held-out TEST summary: accuracy and temporal coherence (all values banked)

scene	P@1.5	R@1.5	P_{pop} OURS	P_{pop} Canny	P_{pop} ratio	ratio (silhouette-ctrl)	Fréchet ratio
chair	0.6573	0.5959	0.067	0.755	11.35×	7.01×	29.92×
lego	0.6196	0.2856	0.063	0.719	11.49×	6.49×	14.03×
ficus	—	—	—	—	—	—	—

P@1.5 / R@1.5: segment-raster, held-out TEST views, stage AFTER pull+prune[tuned+len] (m1b_{scene}_gated_test.json). P_{pop} columns: 240-frame held-out orbit, identical warp operator for both pipelines.
ficus is EXCLUDED by scene scoping, not missing by omission: only 33% of its object pixels lie >4 px from a silhouette, so 'crease vs flat surface' is not well posed (m1b_headline_table.md). No ficus result was ever run.

APPENDIX A
CAN A DISCRIMINATOR PATCH THE BOUNDARY? A PRE-EMPTIVE FALSIFICATION

(All numbers from `RESULTS_MASTER.md` *§4, sourced in* `phase1e_test_eval.json`*,* `phase1e_scores_meta.json`*,* `phase1f_test_eval.json`*, plus banked ledger numbers quoted unchanged; full protocol and caveats in* `PHASE1E_GATE_CHECK.md` */* `PHASE1F_CHAIN_GATE.md`*.)*

The natural reviewer objection to §5 is operational: *"the crease-vs-texture signal exists (0.8401/0.9044) — why not just train a lightweight discriminator, or pool it over chains, and filter the texture edges out?"* We ran that experiment, both ways, before a reviewer could ask — as two post-lock falsification probes under the paper's own discipline: thresholds frozen on VAL views only, TEST evaluated exactly once per arm, and the method-path invariant intact (every gate SCORE is mesh-free at the target scene; the in-scene mesh-supervised probe appears only as a labelled oracle upper bound, never as a result). The candidate set is the Phase-1b chair triangulated cloud (272,366 points), and the metric is the same macro segment-raster P@1.5px convention as the pipeline baseline it must beat (0.6573 at recall 0.5959, quoted from its banked run, never re-run).

A.1 Mesh-free gating converts nothing (the supervision barrier, at the cloud level). The only deployment-legal transfer discriminator at chair — fit on the *other* scene's mesh labels — carries AUC 0.5626 (chance; §5.4's 0.8245 is the *reverse* direction, fit on chair's own mesh, and is therefore not legal as a chair gate). Gated at matched recall, it moves precision from 0.3189 (ungated) to 0.3216. The best in-scene mesh-free gate (the physics-frozen geometric-photometric vote of §5.4) reaches **0.4458** at recall 0.6156 — far under both the 0.6573 baseline and the probes' frozen 0.71 GO bar — and already breaks crease connectivity (topology guard 0.8811, bar 0.90). NO-GO, with margin.

A.2 Even the oracle cannot filter its way through (the topological trilemma). The same cloud gated by an in-scene *mesh-supervised* probe (in-sample AUC 0.9578 — the loosest possible upper bound, and mesh-in-the-loop, so an oracle by construction) reaches P@1.5 **0.7970** at matched recall — and retains only **0.6372** of the baseline-covered GT crease segments: per-point selection drops the low-scoring bridge points that keep 1D chains connected. Chain-mean pooling — the reviewer's fix, run exactly as proposed on the label-pure chain structure of §5.3 (1,692 components) — repairs part of that (guard 0.6372 → 0.7481 mean-pooled, 0.7970 median-pooled) at a real precision cost (point-oracle 0.7970 → chain-gated 0.6458, *below* the 0.6573 baseline), and its ceiling is structural: keeping **every** chain with no thresholding at all already caps the guard at **0.8782 < 0.90**, because the 42.36 % of candidates no proximity+direction chain absorbs are precisely the bridging points. Across every tested gate — mesh-free or oracle, point-gated or chain-pooled — no tested operating point attains precision ≥ 0.71, recall $\geq$ the baseline's 0.5959, and topology ≥ 0.90 *simultaneously*. That is the trilemma, and it is a property of the repair class: post-hoc keep/drop filtering of a fixed candidate cloud trades precision against 1D continuity for every ranking we could construct, up to an in-sample oracle. The oracle also bounds the class: any learned gate's ranking on this fixed cloud is dominated by the in-sample oracle, so the trilemma binds post-hoc discriminator gating as a *class*, not merely the gates we built.

A.3 What this hardens. Two independent barriers now stand between the cloud and a deployable precision fix: the supervision gap (mesh-free discrimination is chance-level where it is legal, §5.4 and A.1) and the continuity severing (even oracle ranking cannot threshold without fragmenting chains, A.2). Fixing either alone is measurably insufficient; a build that wanted this precision would need *topology-aware construction* — connectivity as a constraint during line formation, not a casualty of filtering after it — plus labeled scenes. This is why §5's boundary ships as a measured boundary: the obvious patches are not merely unattempted, they are falsified. Probe scope, disclosed: cloud-level results are chair-only (n=1 scene) — chair being the deliberately selected texture-stress case (Canny edge purity 0.28), so the repair class was falsified on its adversarial home ground, no scene-independent impossibility is claimed, and extending the falsification beyond chair is named future work; the oracle bound is in-sample (loosest); the cloud is rasterised as points inside the segment-raster convention (its 272k points saturate recall at 0.9540 ungated, so the NO-GO is not a rasterisation artifact).

A.4 A learned line field cannot escape the boundary either (the epipolar accumulation test). *(Numbers from* `EPIPOLAR_ACCUM_RESULTS.md`*; script* `scripts/epi_accum.py`*, arrays* `out/epi/`*. Mesh EVAL-ONLY: it supplies the two label sets and the visibility of their points; the accumulated feature is mesh-free.)* The remaining objection is representational: a dedicated 3D edge field, trained on the detector's multi-view 2D edge maps and combined with the frozen 3DGS for occlusion, might integrate sub-threshold evidence coherently across views and so recover creases that per-view thresholding misses. Any such field, whatever its architecture, can learn at most what its multi-view projection loss can see; the closed-form upper bound of that loss is the occlusion-aware multi-view mean $\bar{S}$ of the *raw* detector probability at the true 3D location. We evaluated $\bar{S}$ on the GT creases the frozen pipeline misses (lego 437,138 points, chair 73,371 — the pre-registered miss-set of the coverage analysis) against an equal number of genuinely flat GT surface points (farther than 3 px, twice the scoring tolerance, from any geometric $\geq 10°$ edge, boundary edge, or non-manifold edge of the position-merged mesh; visible in ≥ 1 held-out view under the same mesh-depth rule), over all 100 views, with occlusion from the frozen 3DGS z-buffer under the pipeline's own tolerance, using DexiNed's raw sigmoid maps (never thresholded; continuity asserted at run time) and the pipeline's half-pixel convention, calibrated on the recovered creases. Gate, frozen before any number:

GO iff AUC $\geq$ 0.80 and recall $\geq$ 0.55 at 85 % precision; NO-GO iff AUC $\leq$ 0.65 or that recall $\leq$ 0.42.

Result: **lego AUC 0.698, recall at 85 % precision 0.000; chair 0.859 / 0.000 — NO-GO on both scenes.** The single-view baseline (the same points scored in one view, balanced) reaches AUC 0.644 / 0.738 (best views 0.758 / 0.828); the paired multi-view lift is +0.049 (lego) / +0.133 (chair) AUC — real, but it does not reach the recall criterion. Two readings the numbers rule out. (i) *Dead zeros*: only 11 % of lego's missed creases are undetected by the pipeline's own rule in every visible view (they are true dead zeros, $\bar{S}$ median 0.010); the other 89 % are detected somewhere and score AUC 0.76 against flat surface. (ii) *An over-strict negative class*: the flat set's distance to the nearest geometric edge sweeps the criterion continuously — $\bar{S}$ on flat surface is 0.136 at 3–4 px, 0.097 at 4–5 px, 0.073 at 5–6 px, 0.036 at 8–12 px (missed creases 0.220) — i.e. the detector's response tail reaches $\approx$5 px, and 83 % of lego's flat surface lies within 4.5 px of an edge (97 % within 6 px). Widening the margin lets the criterion pass (lego 6 px: AUC 0.831, recall 0.665) only by restricting the negative class to the 3 % of the surface that is farther from any edge than the detector's footprint — which is not the surface a line field must reject. On chair the zero recall is a knife-edge (removing the 1,000 highest-scoring flat points — 1.4 %, on rails that are silhouettes in most views — lifts it to 0.53); on lego it is structural (removing 5,000 lifts it to 0.055). The kill mechanism is therefore the spatial resolution of the 2D supervision on an edge-dense surface, not missing signal, and it binds any field trained on that supervision. All mesh-free analysis choices (native/multi-scale maps, bilinear/1.5 px-max sampling, two occlusion tolerances, 100 or 80 views, mean/median/trimmed/logit/max aggregation, a 3 px silhouette peel) leave lego's best AUC at 0.713 and its recall at 85 % precision below 0.004; the headline numbers were reproduced to 10^{-9} by an independent re-computation from the saved arrays. A first run of this test with a flat set built from the split-vertex adjacency (the topology issue of A.5) read 0.572 / 0.733 and is superseded.

A.5 The ground-truth crease set is a topology-selected subset. The crease set is built from face adjacency of the mesh as loaded from the asset file, and the NeRF-synthetic OBJs store split vertices (one per position/texture/normal triplet): a hard-shaded crease or a UV seam is therefore an *open-boundary* edge with no face adjacency and is absent from the crease set. Measured on position-merged copies of the meshes, the crease set used throughout covers **37 % (lego) / 43 % (chair) / 31 % (ficus)** of the geometric $\geq$30° edges, and none of a plant's open leaf rims; every precision/recall number in this paper — ours and every baseline's — is scored against that subset. The subset does not flatter the ceiling: on the full geometric crease set the frozen multi-view triangulation of §5.2 recovers 0.175 (lego) / 0.598 (chair) of visible creases, with the never-counted creases slightly *harder* (0.159 / 0.545) than the counted ones (0.199 / 0.665), so the coverage conclusions stand; the absolute recall values are definition-specific and are always to be quoted with it.